Write a complete derivation of the proof that the maximization of margin d is equivalent to minimize the norm of the parameter vector in the maximum margin machine presented in the last set of slides of this module.

Solution.
A separating hyperplane can be written as

$$\mathbf{w}^\top \mathbf{x} + b = 0,$$

where $\mathbf{w}$ is the normal vector and b is the bias. For linearly separable data $(\mathbf{x}_i, y_i)$ with $y_i \in \{-1, 1\}$, we scale $\mathbf{w}$ and b so that

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1.$$

This means that the two margin planes are

$$\mathbf{w}^\top \mathbf{x} + b = 1 \quad \text{and} \quad \mathbf{w}^\top \mathbf{x} + b = -1.$$

Let $\mathbf{x}_0$ be a point on the separating hyperplane:

$$\mathbf{w}^\top \mathbf{x}_0 + b = 0.$$

A point on the upper margin can be written as $\mathbf{x}_1 = \mathbf{x}_0 + \rho\mathbf{w}$. Substituting into $\mathbf{w}^\top \mathbf{x}_1 + b = 1$ gives

$$\mathbf{w}^\top (\mathbf{x}_0 + \rho\mathbf{w}) + b = 1 \quad \Rightarrow \quad \rho\|\mathbf{w}\|^2 = 1 \quad \Rightarrow \quad \rho = \frac{1}{\|\mathbf{w}\|^2}.$$

The perpendicular distance from the hyperplane to this margin is

$$d = \|\mathbf{x}_1 - \mathbf{x}_0\| = \rho\|\mathbf{w}\| = \frac{1}{\|\mathbf{w}\|}.$$

Since $d = 1/\|\mathbf{w}\|$, maximizing the margin is the same as minimizing $\|\mathbf{w}\|$. So the maximum–margin problem can be written as

$$\min_{\mathbf{w}, b} \|\mathbf{w}\| \quad \text{subject to} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1.$$

Therefore, maximizing the margin d is equivalent to minimizing the norm of the parameter vector $\mathbf{w}$ in the maximum–margin machine.